The Pharmacology of Acid Suppression in Children. Proton Pump Inhibitor Dosing, CYP2C19, Tolerance to H2 Blockers, Vonoprazan and Prokinetics.

Imagine a nine-year-old girl who weighs 28 kilograms. She had heartburn and epigastric pain, and an upper endoscopy showed erosive oesophagitis, grade B on the Los Angeles scale, meaning mucosal breaks longer than five millimetres that do not join across the tops of the folds. She was started on omeprazole 10 milligrams, taken at bedtime, for eight weeks. A repeat endoscopy now shows no healing. The referring doctor asks whether she has refractory reflux disease and whether it is time for a stronger drug, or even surgery. Before answering, it is worth asking a simpler question: why did the drug fail?

The main idea of this chapter is that acid suppression fails far more often because of pharmacology than because of resistant disease. A proton pump inhibitor is a prodrug that is activated by acid and blocks only those acid pumps that are working at the moment it arrives. That means four things decide how much acid is actually suppressed: the dose per kilogram of body weight, the timing in relation to a meal, the child's genetic make-up for the liver enzyme called CYP2C19, and the formulation and the way it is given. Beyond the proton pump inhibitors, the H2-receptor antagonists lose their effect within weeks, the new potassium-competitive acid blocker vonoprazan is still essentially an adult drug, and prokinetic drugs have almost no place in paediatric reflux.

A few terms first. Proton pump inhibitors, which we will call PPIs, include omeprazole, esomeprazole, lansoprazole, pantoprazole and rabeprazole. H2-receptor antagonists, which we will call H2 blockers, include famotidine, cimetidine and the now-withdrawn ranitidine. Potassium-competitive acid blockers, or P-CABs, are a newer class, of which vonoprazan is the best known. Prokinetics are drugs meant to speed gastric emptying or strengthen the lower oesophageal sphincter, such as metoclopramide, domperidone and erythromycin.

The sources for this chapter are the 2018 joint paediatric reflux guideline from the North American and European societies for paediatric gastroenterology, hepatology and nutrition, known as NASPGHAN and ESPGHAN, led by Rosen; the adult guideline of the American College of Gastroenterology, known as the ACG, from 2022; the 2024 clinical practice update from the American Gastroenterological Association, known as the AGA, on potassium-competitive acid blockers; the 2021 guideline of the Clinical Pharmacogenetics Implementation Consortium, known as CPIC, on CYP2C19 and PPI dosing; the PHALCON-EE trial of vonoprazan published by Laine and colleagues in Gastroenterology in 2023; and a series of studies on tolerance, safety and regulatory decisions that we will meet along the way.

Part one. How a proton pump inhibitor works.

The acid in the stomach is made by the parietal cell, which pumps hydrogen ions into a tiny channel called the secretory canaliculus, in exchange for potassium, using an enzyme called the hydrogen-potassium ATPase, or proton pump. A PPI travels a strange path to reach this pump. It is destroyed by acid, so it is given with an enteric coating that protects it in the stomach. It is absorbed intact from the small intestine and reaches the parietal cell through the bloodstream. Because it is a weak base, it becomes trapped and concentrated, roughly a thousand-fold, in the acidic canaliculus. There, acid converts it to its active form, which binds permanently, by a covalent bond, to the pump.

Three consequences follow, and each one matters at the bedside. First, a PPI can only block pumps that are active and secreting acid at the time. Pumps are recruited when a meal is expected, so the dose should be taken thirty to sixty minutes before the first meal of the day. A dose at bedtime, with no meal afterwards, meets resting pumps and is largely wasted. Second, although the drug stays in the blood for only one to two hours, the block lasts until the cell makes new pumps, which takes about two days. As a result it takes three to five days of daily dosing to reach steady inhibition. Third, pumps made overnight escape a morning dose, which explains the so-called nocturnal acid breakthrough. Splitting the dose, before breakfast and before the evening meal, blocks more pumps than simply doubling a single morning dose.

It helps to compare the four ways of lowering gastric acid. PPIs block the pump permanently but need acid to be activated and take several days to reach full effect; they depend strongly on CYP2C19, more so for omeprazole than for pantoprazole; and they do not lose effect over time. H2 blockers block the histamine signal that drives the pump; they act within hours and do not need acid, but tolerance develops within days to weeks. Vonoprazan blocks the pump reversibly at the potassium binding site, needs no acid activation, works from the first dose, and depends only a little on CYP2C19, being cleared mainly by another enzyme, CYP3A4. Antacids and alginates neutralise acid or form a raft on top of the stomach contents, act within minutes, and are useful only for short-term relief. In children, PPIs remain the workhorse, and every other class is a niche or a bridge.

Part two. Dosing by weight and age.

Children are not small adults, and drug clearance changes with age. Infants under about three months have immature CYP2C19 and CYP3A4 enzymes, so they clear PPIs slowly and their exposure for a given dose is high. Children between about one and six years clear these drugs faster per kilogram than adolescents and adults, so they need relatively more drug. From about 40 kilograms or twelve years of age, adult doses apply.

Labelled doses are given in weight bands, and the bands hide a two-fold range. Take omeprazole for children aged one to sixteen years: 5 milligrams for a child of 5 to just under 10 kilograms, 10 milligrams for 10 to just under 20 kilograms, and 20 milligrams from 20 kilograms upwards. A child who weighs 5 kilograms receives 1 milligram per kilogram, but a child weighing 9.9 kilograms, still in the same band, receives only half a milligram per

kilogram. The same happens at the top of each band: 10 milligrams at 19.9 kilograms is half a milligram per kilogram, and 20 milligrams at 40 kilograms is the same. For erosive disease, it is reasonable to round up to the next band or to split the dose twice daily, rather than accept half a milligram per kilogram.

The other labelled doses follow the same logic. Esomeprazole is licensed for erosive oesophagitis in infants from one month of age, at 2.5, 5 or 10 milligrams by weight, then at 10 milligrams for children under 20 kilograms and 10 or 20 milligrams above that, and 20 to 40 milligrams in adolescents. Lansoprazole is given at 15 milligrams up to 30 kilograms and 30 milligrams above. Pantoprazole is licensed from five years, at 20 milligrams up to 40 kilograms and 40 milligrams above, and it depends least on CYP2C19. Famotidine is dosed at half a milligram per kilogram once daily under three months of age, half a milligram per kilogram twice daily up to one year, and about 1 milligram per kilogram per day in two doses in older children, up to 40 milligrams twice daily, with a lower dose in kidney impairment.

One point needs emphasis. No PPI is licensed for crying, irritability or regurgitation in infants. In the multicentre randomised trial of lansoprazole in 162 infants aged one to twelve months, published by Orenstein and colleagues in 2009, 54 percent of infants responded on lansoprazole and exactly 54 percent responded on placebo. Serious adverse events, mostly lower respiratory tract infections, occurred in ten infants on lansoprazole and in two on placebo. Acid is rarely the cause of a crying baby.

Part three. Timing and formulation.

Before escalating any dose, ask how, when and with what the dose is taken. A dose taken thirty to sixty minutes before breakfast works; a bedtime dose is the least effective. The enteric-coated granules must not be crushed or chewed; capsules can be opened and the granules sprinkled onto a spoonful of soft, slightly acidic food such as apple purée. Granules can clog small feeding tubes, so an orodispersible tablet, such as lansoprazole, or a suspension is better for gastrostomy tubes. In a child on continuous tube feeds there is no meal signal to recruit pumps, so the dose is best given during a break in the feed, and twice-daily dosing may be needed. Compounded liquid omeprazole, made up in sodium bicarbonate, has a short shelf life and variable content, so it is worth checking the pharmacy recipe. Correcting timing and formulation is cheaper and safer than doubling the dose, and it is the step most often skipped.

Part four. CYP2C19 and the genetics of exposure.

Omeprazole, lansoprazole and pantoprazole are cleared mainly by the liver enzyme CYP2C19, and the gene for this enzyme varies between people. The most common loss-of-function variant is called star two. A gain-of-function variant called star seventeen makes the enzyme work faster. A child with two star seventeen copies is an ultrarapid metabolizer and has the lowest drug exposure. One star seventeen copy with a normal copy makes a rapid metabolizer. Two normal copies make a normal metabolizer. One star two copy makes an intermediate metabolizer, and two star two copies a poor metabolizer, with the highest exposure.

The CPIC guideline, published by Lima and colleagues in 2021, translates this into dosing. For ultrarapid metabolizers, it recommends increasing the starting daily dose by 100 percent, that is doubling it, and dividing it if needed. For rapid and normal metabolizers, the usual starting dose is used, with an increase of 50 to 100 percent considered for erosive oesophagitis or Helicobacter pylori infection. For intermediate and poor metabolizers, the usual starting dose is used, but after more than twelve weeks of successful treatment a 50 percent dose reduction should be considered, because exposure is high and long-term side effects may be more likely. The guideline states that these recommendations apply to children, but not to neonates and infants under about two to three months, whose clearance is very low regardless of genotype.

How common are these variants? The star two allele is found in about 15 percent of people of European or African ancestry, 25 to 30 percent of Asian ancestry, and about 60 percent of people from Oceania. Another loss-of-function allele, star three, is found in 2 to 7 percent of Asian people. The star seventeen allele is found in about 20 percent of people of African, European and Near Eastern ancestry. Put simply, in a European or African clinic about one child in three carries a star seventeen allele and clears PPIs faster, whereas in East Asia poor metabolizers are common, which is one reason why PPI trials from Asia often showed excellent results.

In children, rapid and ultrarapid metabolizer status has been linked with poorer response to PPIs in reflux disease and in eosinophilic oesophagitis, in cohorts reported by Franciosi, Mougey and colleagues; poor metabolizers taking lansoprazole had more respiratory infections; and a pilot pragmatic trial of genotype-guided dosing in children, led by Cicali, showed that the approach is feasible. In practice, genotype rarely changes the first prescription, but it can explain failure at a full dose and support stepping down during long-term treatment.

Part five. H2 blockers and tolerance.

H2 blockers work quickly and do not need acid, but their effect fades. In a classic study published by Nwokolo and colleagues in 1990, 48 healthy adults took standard doses of cimetidine, nizatidine, famotidine or ranitidine for 29 days while their stomach acidity was measured overnight. The chart from this study is striking. Without any drug, the median overnight acidity was 485 units. On the first night of treatment it fell to 35 units, a reduction of about 93 percent. By day fifteen it had risen to 67, and by day twenty-nine to 117, which is about a quarter of the untreated level and more than three times the level on the first night. All four drugs behaved alike, and the tolerance developed while gastrin, the hormone that drives acid secretion, remained high. Tolerance cannot be overcome simply by raising the dose.

This shapes how H2 blockers are used. They are reasonable for on-demand or short courses, or when a PPI is unavailable or not tolerated. Adding a bedtime H2 blocker to treat night-time breakthrough on a PPI is a common habit, but its effect fades within weeks, so twice-daily PPI dosing is usually a better answer. Ranitidine was withdrawn from the market in 2020 because of contamination with a probable carcinogen called N-nitrosodimethylamine, and famotidine is

now the main choice.

Part six. Vonoprazan, the potassium-competitive acid blocker.

Vonoprazan works differently. It is stable in acid, so it needs no enteric coating and can be taken regardless of meals. It is a strong base and concentrates in the canaliculus whether the pumps are active or resting. It blocks the pump by competing with potassium at its binding site, reversibly, and it does not need to be activated. Its effect is close to maximal from the first dose, and its half-life of about seven hours means that it covers the night. It is cleared mainly by CYP3A4, so CYP2C19 genotype matters little. In other words, vonoprazan removes the three classic reasons for PPI failure: timing, night-time breakthrough and rapid metabolism. The price is a larger rise in gastrin and less long-term safety information, particularly in children.

The key evidence comes from the PHALCON-EE trial, which randomised 1,024 adults with erosive oesophagitis to vonoprazan 20 milligrams or lansoprazole 30 milligrams. At eight weeks, 92.9 percent of patients on vonoprazan were healed compared with 84.6 percent on lansoprazole. The difference was largest in severe disease: among patients with Los Angeles grade C or D oesophagitis, about 70 percent were healed after only two weeks on vonoprazan, compared with about 53 percent on lansoprazole. Healing was maintained at twenty-four weeks in about 81 percent on vonoprazan 20 milligrams and 79 percent on 10 milligrams, compared with 72 percent on lansoprazole 15 milligrams. All these differences were statistically significant. The message is that vonoprazan's advantage is greatest where acid control matters most, in severe oesophagitis and in the first two weeks; in milder disease the margin over a well-taken PPI is small.

Vonoprazan has been approved in Japan since 2015 and in the United States since 2023 for adults with erosive oesophagitis. The 2024 AGA update recommends potassium-competitive acid blockers for severe, grade C or D, oesophagitis, for patients with proven reflux disease who fail twice-daily PPI therapy, and as part of Helicobacter pylori eradication, but not as the routine first choice for every patient with heartburn. In children, only pharmacokinetic studies have been completed: two studies, reported by Mulford and colleagues in 2026, gave 10 or 20 milligrams daily for fourteen days to 42 children aged six to seventeen. There is no paediatric efficacy trial, so any use in children is off-label and should be confined to severe, refractory disease, after counselling.

Part seven. Prokinetics: why almost none.

The idea behind prokinetics is attractive, but it rests on a misunderstanding. Reflux in children is driven mainly by transient relaxations of the lower oesophageal sphincter, not by slow gastric emptying, so drugs that speed emptying have little to act on. Meanwhile, each available prokinetic carries a safety signal larger than its benefit.

Metoclopramide carries a boxed warning from the United States Food and Drug Administration, issued in 2009, for tardive dyskinesia, a potentially permanent movement disorder. In 2013 the European Medicines Agency, known as the EMA, contraindicated it in children under one year, limited treatment to five days, set the paediatric dose at 0.1 to 0.15 milligrams per kilogram up to three times a day, and stated that it should not be used for chronic conditions such as reflux. Domperidone was restricted by European regulators in 2014 to nausea and vomiting only, at 0.25 milligrams per kilogram up to three times a day for children under 35 kilograms, for no more than a week, because of prolongation of the QT interval and the risk of dangerous heart rhythms; it is not approved in the United States. Cisapride was withdrawn in 2000 after fatal arrhythmias. Baclofen, a GABA-B receptor agonist, reduces transient relaxations of the sphincter, but causes sedation and can lower the seizure threshold; the paediatric guideline allows it to be considered before surgery, in specialist hands.

Erythromycin deserves special mention. At low doses it stimulates the motilin receptor and speeds gastric emptying, and it is sometimes used for feeding intolerance in neonatal units. But the same effect drives hypertrophy of the pylorus in young infants. In a large cohort from the United States military health system, reported by Eberly and colleagues in Pediatrics in 2015, exposure to erythromycin in the first fourteen days of life was associated with odds of pyloric stenosis about thirteen times higher, and azithromycin about eight times higher. Between days fifteen and forty-two, the odds were about four times higher with erythromycin and three times higher with azithromycin. After six weeks there was no association. Erythromycin should therefore be avoided as a prokinetic in the first six weeks of life, and parents should be warned if it is used for infection.

Part eight. Safety and interactions.

Acid suppression in infancy is not free. In a cohort of 792,130 children, reported by Mitre and colleagues in JAMA Pediatrics in 2018, children who received acid suppression in the first six months of life had a higher rate of later food allergy, with a hazard ratio of 2.59 for PPIs and 2.18 for H2 blockers, and a higher rate of anaphylaxis, with hazard ratios of about 1.45 and 1.5. In a cohort of 851,631 children, reported by Malchodi and colleagues in Pediatrics in 2019, PPI use in the first year was associated with a 21 percent higher risk of later fracture, and combined PPI and H2 blocker use with a 30 percent higher risk. These are observational associations and may be partly explained by the reasons for treatment, but they are consistent, related to dose and duration, and seen with both classes. Signals for gastroenteritis, pneumonia, Clostridioides difficile infection, and necrotising enterocolitis in preterm infants given H2 blockers point the same way. The conclusion is not to withhold treatment for real oesophagitis, but to give short, indicated courses and to review every course at eight to twelve weeks.

Interactions matter especially in transplant and inflammatory bowel disease clinics. Omeprazole and lansoprazole can raise tacrolimus levels, most of all in CYP2C19 poor metabolizers, so pantoprazole or famotidine is preferred and the tacrolimus trough should be

checked after starting or stopping acid suppression. A higher gastric pH reduces the absorption of mycophenolate mofetil; the enteric-coated mycophenolate sodium is less affected. Long courses reduce absorption of iron and vitamin B12, so ferritin and B12 should be checked. Some antifungals, including itraconazole, posaconazole suspension and ketoconazole, need acid for absorption. Omeprazole blocks the activation of clopidogrel, which matters in adolescents and adults. And PPIs have been reported to delay the clearance of high-dose methotrexate, so they are usually held around such infusions.

Part nine. What the guidelines say.

On the first-line treatment of erosive oesophagitis, the documents agree: the 2018 paediatric guideline recommends a PPI for four to eight weeks, the 2022 ACG guideline advises taking the PPI thirty to sixty minutes before a meal, and the CPIC guideline adds that genotype may justify a higher dose. They differ on H2 blockers: the paediatric guideline reserves them for when a PPI is unavailable or contraindicated, whereas the adult guidance allows a bedtime H2 blocker for night-time symptoms while acknowledging tolerance. On prokinetics they agree: the paediatric guideline advises against them as first-line treatment, allowing baclofen to be considered before surgery, and the adult guideline advises against them unless there is gastroparesis. The material difference concerns vonoprazan: the paediatric guideline pre-dates its approval and does not address it, while the 2024 AGA update supports it for severe oesophagitis, PPI failure and Helicobacter pylori. On genotype-guided dosing, CPIC states that its recommendations apply to children over about three months, but neither society guideline addresses it. And on duration, all agree on the lowest effective dose for the shortest time, with a trial of four to eight weeks followed by weaning in children.

Part ten. Traps and pearls.

Several traps recur. Doubling the dose before fixing timing and formulation. Adding ranitidine at night, which is no longer available, or relying on any H2 blocker long term despite tolerance. Giving a PPI to the crying baby, when placebo works just as well and infections increase. Reaching for a prokinetic when reflux is caused by transient sphincter relaxations and the drugs are restricted for safety reasons. Labelling PPI failure as refractory reflux without checking adherence, considering genotype, and taking biopsies for eosinophilic oesophagitis. And forgetting to stop, so that a four-to-eight-week trial becomes years of therapy.

Return now to the nine-year-old girl. At 28 kilograms, 10 milligrams of omeprazole was about 0.36 milligrams per kilogram, below the labelled 20 milligrams for her weight, and it was taken at bedtime, when her pumps were idle. The first step is 20 milligrams thirty minutes before breakfast, or the same total split before breakfast and dinner, with a repeat endoscopy after eight to twelve weeks. If she fails at a full, properly timed dose, check adherence, consider CYP2C19 genotyping, and make sure biopsies have excluded eosinophilic oesophagitis. If she were a liver transplant recipient on tacrolimus, pantoprazole or famotidine would be preferred, with a fresh tacrolimus trough. An adolescent with grade D oesophagitis

refractory to twice-daily PPI could be considered, by a specialist and after counselling, for off-label vonoprazan. An infant on a PPI for crying for six months should simply stop. And a three-week-old who is vomiting should not be given erythromycin as a prokinetic.

So, tomorrow in clinic, carry these points. A PPI blocks only active pumps: give it thirty to sixty minutes before breakfast, and expect full effect after three to five days. Weight bands give between half a milligram and one milligram of omeprazole per kilogram, and children aged one to six need relatively more. No PPI is licensed for infant crying, and in the lansoprazole trial response was 54 percent with drug and with placebo, with ten serious respiratory infections against two. Ultrarapid CYP2C19 metabolizers need a doubled starting dose, and poor metabolizers can have their dose halved after twelve weeks of control; about one in three European children carries the fast star seventeen allele. H2 blockers cut night-time acidity by 93 percent on the first day, but the acidity climbs more than three-fold again within a month. Vonoprazan healed about 93 percent against 85 percent of adults with erosive oesophagitis at eight weeks, and about 70 against 53 percent of severe cases at two weeks, but it remains unproven in children. Metoclopramide is not for children under one year or for more than five days, and domperidone is limited to a week at a quarter of a milligram per kilogram up to three times daily. Erythromycin in the first two weeks of life multiplies the odds of pyloric stenosis about thirteen-fold. And acid suppression in early infancy is associated with more food allergy and more fractures, which is the best reason to prescribe it only for a clear indication, at the right dose and time, and for as short a course as possible.